

Retail Performance Analysis Report

Prepared by: Keerthana Madarapu | June 2024

Executive Summary

This report analyzes 9,994 retail transactions across 4 regions of the United States. The analysis was conducted using Excel, SQL, and Power BI to identify key performance trends, profitable segments, and areas requiring immediate business attention.

Key Findings

1. The West region leads in sales with \$725,457, representing 32% of the total company revenue of \$2.3 Million.
2. Technology is the most profitable category, generating \$145,454 in profit despite Furniture having a similar sales volume of \$741,999 but only \$18,451 in profit.
3. Texas is the highest loss-making state with -\$25,729 in profit, followed by Ohio at -\$16,971 and Pennsylvania at -\$15,559.
4. Canon image CLASS 2200 Advanced Copier is the top-selling product, generating \$61,599 in sales — almost double the second-best-selling product.
5. The South region is underperforming with only \$391,721 in sales compared to the West region at \$725,457 — a difference of 46%.

Business Recommendations

1. Investigate Furniture Profitability. Furniture generates \$741,999 in sales but only \$18,451 in profit — a profit margin of just 2.5%. The business should review discount strategies and supplier costs in the Furniture category to improve margins.
2. Address Loss-Making States: Texas, Ohio, and Pennsylvania are collectively losing \$58,260 in profit. The business should conduct a regional pricing and discount audit in these states to identify root causes and reduce losses.

3. Expand the West Region Strategy to the South Region. The West region strategy is clearly working with \$725,457 in sales. The business should analyze what is working in the West region and apply the same strategies to the underperforming South region to close the 46% sales gap.

Tools and Methodology

Data Source: Superstore Retail Dataset (9,994 transactions)

Tools Used:

- Microsoft Excel: Data cleaning, formatting, and Pivot Table analysis
- SQL (SQLite): Data querying and business questions analysis
- Power BI: Interactive dashboard and data visualization

Analysis Steps:

1. Data was cleaned and formatted in Microsoft Excel
2. Three Pivot Tables were created to summarize key metrics
3. Five SQL queries were written to answer business questions
4. An interactive Power BI dashboard was built with 5 visuals